American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
3	2/4/20	9 am	30 mins	◉ Advisor ○ Students	Hadi Elham

Attendees:

Lynn Ariss, Raghad Al AGha, Claritta Khoury, Prof. Ayman Kayssi

Briefly summarize the main discussions during the meeting:

The meeting focused on resolving inconsistencies in the machine-learning pipeline, particularly the mismatch between 30-second and 5-second data windows, which affected label correctness and model performance. The team discussed restructuring the dataset into proper 5-second segments aligned with the reference methodology, validating the new labels, and improving training results. Parallel efforts included testing real-time model behavior, integrating outputs into an application, and preparing IRB-compliant data storage using anonymized CSV files. On the hardware side, discussions covered IMU-based step counting, pressure-sensor matrix design challenges (including resistance cross-talk and multiplexing), battery and charging limitations, voltage regulation requirements, and practical constraints in sourcing components locally. Design preferences for a shoe-based configuration and upcoming deadlines were also addressed.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that rebuilding the dataset at a true 5-second resolution significantly improved model reliability and should serve as the new baseline, followed by fine-tuning and real-time testing. Data handling will comply with IRB requirements through anonymized, timestamped storage, while app integration will proceed in parallel. Hardware development will prioritize a stable shoe-based design, cautious expansion of the pressure-sensor matrix, and a safe, regulated power system despite local sourcing limitations. Overall, the project is moving from exploratory experimentation toward a more structured, deployable, and clinically aligned implementation.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Work on the charging mechanism addressing discussed issues, continue heatmap generation, translate the IRB consent form, read papers about accurate ranges for correlating flex sensor deviations to edema severity.	Raghad Al Agha	2/11/2026
Try training configurations for ataxia detection, fix heartrate retrieval with app integration, encrypt arduino communication, aid in heatmap generation and flex sensor configuration for edema detection.	Hadi Elham	2/11/2026
enhance the app's security by implementing password and biometric sign-in, integrate CSV file handling for IRB data storage, and perform hyperparameter tuning to optimize the model's performance.	Lynn Ariss	2/11/2026
Finalize IRB proposal, consent forms, questionnaires, Google forms, flyers, user manual, and coordinate w/ Prof. Massoud to actually submit	Claritta Khoury	2/11/2026